

Project Work - AIN

Deep Learning and Big Data

1 General Information

Students will work in groups on a project throughout the semester. The goal is to understand, implement, and critically compare deep learning models.

2 Project Timeline

- **Group Formation (April 16, 2026 – April 23, 2026)**
 - Groups of 4 students (AIN)
- **Dataset Selection (April 23, 2026 – April 30, 2026)**
 - Selection of a dataset and project topic
 - Preparation of the project idea
- After selecting the dataset, the project can be adjusted within one month.
- **Submission Deadline (June 23, 2026, 23:59)**
 - Strict deadline
 - Late submissions will not be accepted
 - Automatic grade of **0%**
- **Presentations**
 - Dates:
 - June 25, 2026
 - July 2, 2026
 - July 9, 2026
 - Due to the number of groups, presentations will be held across three sessions
 - Groups select their presentation date after choosing the dataset
 - Attendance at **all presentation sessions is mandatory**
 - Missing attendance will negatively affect the grade
 - 15 minutes presentation + 10 minutes Q&A

3 Project Requirements

- **Core Requirements**

- **Two Models**
 - Model 1: From Scratch
 - Full implementation and training from scratch
 - Justification of all design decisions: architecture, layers, activation functions, loss function, optimizer, and hyperparameters
 - Model 2: Alternative Approach
 - Transfer learning (e.g., for image data)
 - Pretrained models (e.g., NLP)
 - Strong baseline (e.g., classical ML methods)
- **Two Tasks on the Same Dataset**
 - Multi-class classification + binary classification
 - Classification + regression
 - Classification + anomaly detection
 - Classification + time series forecasting
- **Important**
 - The comparison between the models must be clearly included in the report

4 Deliverables

- **Code**
 - Python (PyTorch recommended)
 - Jupyter Notebook, GitHub, or Colab
- **Report (6 – 10 pages)**
 - Problem statement
 - Dataset description
 - Model 1
 - Model 2
 - Results
 - Comparison
 - Discussion and limitations
- **Presentation**
 - Slides in PDF format
 - Clear structure: introduction, methods, results, discussion

5 Grading

5.1 Overall Grading

- **Project (100%)**
 - Coding / Modeling: 40%

- Report: 30%
- Presentation: 30%

Note: The final grade is entirely based on the project.

5.2 Evaluation Criteria

- **Coding / Modeling**
 - Correct implementation and functioning training process
 - Meaningful model architecture
 - Appropriate choice of loss function and optimizer
 - Well-justified hyperparameters
 - Clean structure and readability
 - Reproducibility
 - Independent experimentation
- **Report**
 - Clear problem statement and dataset description
 - Explanation of the models
 - Justification of design decisions
 - Structured presentation of results
 - Interpretation and comparison
- **Presentation**
 - Clear and understandable delivery
 - Well-structured content
 - Ability to answer questions
 - Time management

6 Important Rules

- Plagiarism is strictly prohibited
- External solutions or tutorials must not be copied
- All sources must be properly cited
- Individual contribution must be clearly visible

Violations will result in:

- Grade: **fail**

7 Recommended Datasets

- Food-101: https://data.vision.ee.ethz.ch/cvl/datasets_extra/food-101/
- EuroSAT: <https://github.com/phelber/eurosat>

- Oxford Pets: <https://www.robots.ox.ac.uk/~vgg/data/pets/>
- UCI HAR: <https://archive.ics.uci.edu/dataset/240/human+activity+recognition+using+smartphones>
- Electricity Load: <https://archive.ics.uci.edu/ml/datasets/ElectricityLoadDiagrams20112014>
- PhysioNet: <https://physionet.org/content/respiratory-oximetry-apnoea/1.0.0/>
- AG News: <https://www.kaggle.com/datasets/amananandrai/ag-news-classification-dataset>
- IMDb: <https://ai.stanford.edu/~amaas/data/sentiment/>
- Jigsaw: <https://www.kaggle.com/c/jigsaw-toxic-comment-classification-challenge>
- UrbanSound8K: <https://urbansounddataset.weebly.com/urbansound8k.html>

Note: Other datasets are allowed upon approval.

8 Final Note

Start working immediately after group formation. An early start is essential for project success.